



Enteral Feeding and Enteral Tubes Management – Adult procedure

1. Applicability

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All areas	Medical Nursing Allied Health

2. Purpose

This procedure establishes minimum procedural standards for Percutaneous and Enteral Feeding in adults throughout Armadale Kalamunda Group.

3. Pre-Insertion Risk Assessment

- Contraindications and/ or complications identified should be discussed with the MO and Dietetic staff.
 - Determine the purpose of the tube and liaise with senior staff to determine tube sizing.
 - As a guide for adults: 8-12 FG (French gauge)
 - Determine the last oral intake.
 - Check patient allergies and sensitivities
- Insertion of an NGT/NJT should be ordered by the Medical Officer (MO)
 - Discuss the enteral feeding plan with the patient and/or family and obtain appropriate consent.
 - Staff inserting the NGT/NJT must have the appropriate knowledge and expertise.
 - Patients who have undergone 3 failed attempts at NGT/NJT insertion must be referred to another staff member experienced in this procedure.
 - NGT/NJT are not recommended for facial or base of skull fractures (OG insertion may be considered).
 - NJT's are inserted under radiology guidance.

3.1 Insertion and re-insertion

- Insertion and re-insertion in the following cases should be done by the MO and ONLY in consultation with the registrar of the treating medical team:
 - Maxillofacial surgery / neoplasm
 - Oesophageal/ nasopharyngeal tumours, strictures, fistulae, surgery or varices
 - Coagulopathies
 - Gastric surgery or trauma

- Post upper GI surgery
- Laryngeal surgery

Liaise with Medical Officer if the following are identified:

- Gastric reflux
- Hiatus herniation
- Violent coughing
- Recurrent or violent vomiting/ retching
- Risk for refeeding syndrome

4. Procedure – Nasogastric Tube

4.1 Equipment required

- Appropriately sized NGT – single / dual lumen
- Scissors
- Lubricant
- Torch
- Tissues
- PPE
- Disposable receptacle
- Waterproof protective sheet
- 20/50mL enteral syringe as appropriate
- Emesis receptacle
- pH indicator strips
- Tongue depressor
- Securing tape

4.2 Optional equipment

- Drainage bag
- Skin preparation wipe
- Glass of water and a straw if not nil by mouth or on thickened fluids
- Cap / spigot indelible pen

4.3 Procedure

- Ensure patient privacy.
- Explain the reasons for NGT insertion and the process involved to the patient obtain appropriate consent.
- Examine nostrils for deformity/ obstructions and determine the best side for insertion.
- Position patient in an upright position (30-45 degrees) as clinically indicated. Some neurological or spinal patients will be positioned according to MO instructions.
- Place the waterproof protective sheet over the chest and the receptacle within reach.

- Measure the selected tube length (cm) from nose tip to ear lobe, then bending the tube from the ear lobe to xiphoid process of sternum.
- Weighted silastic tubes should be measured from the end of the weight not the tip.
- Note the required length by reading the markings on the tube. If no markings, mark the position with tape or pen.
- Lubricate the tube according to manufacturer's instructions.
- Place the NGT in a clean receptacle or other appropriate clean surface.
- Insert the lubricated tube tip into the selected nostril, easing it along the floor of the nasal passage toward the ear closer to the chosen nostril.
- Advance the tube slowly.
- Stop if resistance is felt and adjust the direction slightly before proceeding. If resistance persists, remove the tube and attempt via the other nare.
- Ask the patient to take slow even breaths throughout insertion if able to cooperate.
- As the tube passes down into the nasopharynx, ask the patient to start swallowing repeatedly, while tipping the chin downward.
- If the patient is not nil by mouth and can safely swallow, this process is aided by sipping water through a straw.
- Advance the NGT with each swallow, as appropriate to the patient's swallowing ability.
- Inspect the oral cavity using a torch to ensure the tube has not coiled in the mouth.
- If patient gags excessively, withdraw tube slightly and check it is not coiled at the back of the throat.
- For unconscious patients, passage of the tube into the oesophagus may be difficult. Liaise with staff experienced in this procedure.
- Continue advancing the tube until the required marking has reached the entry into the nares.
- Withdraw the tube immediately if changes in respiratory status or speech are observed.
- Secure to the nose and/or cheek keeping clear of the visual field.
- Attach enteral syringe (purple) to the free end of the tube and aspirate sample of gastric contents.
- Confirm tube placement. Refer to [Checking Tube Placement](#) and [Appendix 1 Algorithm for Confirming the Correct Position of OGT/NGT's in Adults](#).

4.4 Post-Insertion care

- Measure and document external tube length from the nares to the hub/tube end.
- Once the tube position is confirmed, secure the tube and manage as clinically indicated.
- Tube securing tapes and skin around the insertion site should be checked at the time of insertion and at a minimum of once per shift.
- The insertion site should be cleansed with water and skin emollient applied once per shift or as clinically indicated.
- Perform mouth care every 2-4 hours, record on nursing care plan.

5. Securing the nasogastric / nasojejunal tube

5.1 Equipment required

- Skin preparation wipe
- Adhesive removal wipe
- Tape measure
- Hydrocolloid dressing (optional)
- Securing tape or fit for purpose product

5.2 If using a fit for purpose securing product

- Attach as per the manufacture instructions (e.g nasal bridle)

5.3 If using Adhesive tape

- Cut a strip of tape the same width and twice as long as the length of the nose.
- Split one end down the middle leaving one third intact.
- Clean outer aspect of nose with an alcohol wipe.
- Apply skin preparation wipe to nose area.
- Attach the uncut section of tape to the patient's nose and secure one side of the split end of tape lengthwise around the tube.
- Wrap the other end around the tube in the opposite direction to the first.
- Secure a second piece of tape over the first, across the bridge of the nose.
- NGT/NJT may be secured to the cheek avoiding the patient's line of vision.
- Consider applying pliable hydrocolloid dressing.

5.4 Removing old tapes

- Change as necessary when there are signs of skin irritation and when the securing tape/product is loose, soiled or due for replacement as per manufacturer's instruction.
- Consider assistance to secure NGT/NJT while old tapes are removed.
- Remove old tapes while maintaining the required external NGT/NJT length.
- Apply adhesive removal wipe to areas with residual tape adhesive.

6. Checking placement

Nasally and orally inserted gastric tubes only, refer to [Appendix 1: Algorithm for Confirming the Correct Position of OGT/NGT's in Adults](#).

- Confirmation of initial placement for naso-gastric and oro-gastric tubes is by an x- ray.
- Liaise with MO for review of the x-ray to confirm correct placement prior to use. MO to document confirmation of tube placement following x-ray
- If the x-ray shows incorrect positioning of the gastric or intestinal tube do not use it.
- Subsequent confirmations of placement are by an aspirate pH of less than 5.5 and measurement of the distance between the nare and the end of tube. Tube length should correspond to the length documented at insertion. Check that tubing is not coiled in the mouth.

- If unsure of placement do not use the tube for any purpose. Consult with Senior Nursing staff, MO and treating medical team.
- Tube placement should be confirmed every shift, prior to every use and whenever there is any doubt as to its position. For example:
 - Following insertion.
 - Once every 24 hours or as per MO instruction if not being used to administer feeds/medications.
 - Prior to each bolus or intermittent feed.
 - Following a break in continuous feeding.
 - Prior to administration of medications.
 - If the external length of the tube has changed.
 - After oro-pharyngeal suction.
 - Following a coughing fit.
 - After vomiting.
 - When the patient complains of discomfort or feed reflux.
 - In the event of sudden respiratory difficulties.
 - When a patient is transferred to another clinical area/site.
- Ensure at least 1 hour since feeding before pH testing. Consider acidity of medicines and other fluids administered.
- Auscultation and visual inspection are not recommended methods of checking.

6.1 Intestinal Tubes e.g. NJT

- Initial placement is confirmed radiologically for naso-duodenal/naso-jejunal tubes.
 - **Do not** aspirate intestinal tubes.
- If unsure of placement do not use the tube for any purpose. Consult with Senior Nursing staff, MO and treating medical team.

6.2 Gastronomy tubes e.g PEG, RIG

- Initial placement is confirmed on insertion endoscopically or radiologically and should be re-confirmed radiologically if in doubt.
- If unsure of placement do not use the tube for any purpose. Consult with Senior Nursing staff, MO and treating medical team.

7. pH Testing

7.1 Equipment Required

- pH indicator strips (non-bleeding)
- Disposable receptacle e.g. kidney dish
- 50mL enteral syringe
- Plastic backed absorbent sheet

7.2 Procedure

- Explain procedure to patient
- Ensure patient privacy

- Arrange plastic backed absorbent pad under the tube port
- Open the cap/spigot on the enteral tube and attach enteral syringe
- Aspirate 0.5-1mL of fluid
- Disconnect the enteral syringe and immediately replace cap/spigot
- Place a few drops of aspirate onto the pH indicator strip and wait 10 seconds
- Match the colour change of the strip with the colour code on the box to identify the pH of the aspirate
- pH less than 5.5 indicates an acid reaction and that the tube is probably placed in the stomach.
- If pH is 5.5 or greater liaise with MO or Pharmacist (potential medication effects e.g. pantoprazole).

8. Nasogastric tube removal

Confirm that the NGT is to be removed with MO, Speech Pathologist and Dietitian as appropriate.

8.1 Equipment Required

- 10mL enteral syringe
- Adhesive removal wipe
- Tissues
- Paper bag
- Waterproof protective sheet
- Emesis bowl
- Cap / spigot

8.2 Procedure

- Ensure patient privacy.
- Explain the procedure to the patient.
- Agree on a signal to indicate the patient wishes to stop the procedure.
- Position the patient 30-45 degrees head up as clinically indicated.
- Place the waterproof protective sheet over the chest.
- Disconnect the drainage bag or feeding administration set as appropriate.
- Remove the securing tape.
- Gently push 10-20mL of air into the NGT.
- Cap/spigot the NGT or kink it by folding in half.
- Ask the patient to take a deep breath while the tube is removed as clinically indicated.
- Remove the NGT slowly and evenly over 3 to 6 seconds.
- Clean the nares with tissue and remove residual skin adhesive with adhesive removal wipes.
- Provide mouth care.
- Monitor the patient over 48 hours following removal of the NGT for:

- Nausea and vomiting
- Abdominal distension
- Tolerance of food

9. Aspiration and management of Gastric Residual Volume

Gastrostomy and intestinal tubes e.g., PEG, RIG and NJT should not be aspirated.

- Gastric fluid aspiration is used to:
 - check placement of the enteral tube on insertion and prior to feeding
 - check gastric residual volume
 - reduce vomiting

9.1 Equipment Required

- Waterproof absorbent sheet
- 50mL enteral syringe
- Receptacle for gastric contents
- pH test strips
- Water – Refer to Feed Administration
 - **Optional:** Flushing equipment if feed in progress: Cap / Spigot drainage bag

9.2 Procedure

- Explain procedure to patient.
- Place plastic backed absorbent sheet under the enteral tube.
- If a feed is in progress - stop the feed and flush the tube with at least 30mL of water.
- Connect 50mL enteral syringe to the enteral tube and gently aspirate the tube.
- Disconnect the syringe and occlude the tube with cap/spigot whilst placing gastric contents into the disposable receptacle.
- Test the pH of the aspirate.
- Return aspirates as required.
- Flush the enteral tube with water.
- Disconnect the syringe and ensure the end of the enteral tube is closed to prevent tube leakage or commence feeding if appropriate.
- If there is no aspirate from an NGT, try one or more of the following:
 - Position the patient onto their side
 - Advance the NGT 2-5cm
 - Inject 10-20mL of air using a 50mL syringe
 - Wait 15-30 minutes

9.3 Gastric Residual Volume (GRV)

Is routinely measured in some clinical areas (not ICU) to monitor:

- Gastric tolerance to intermittent or bolus enteral feeding.

- Abdominal decompression and drainage for patients not receiving enteral feeding.
- Volumes below are a guide only and individual patient characteristics and enteral feeding regimens should be taken into account.
- Refer to departmental specific procedures as required, e.g., [Appendix 3 Intensive Care Unit After-Hours Enteral Feeding Procedure](#).

• Less than 200mL	• Liaise with MO to consider medications to enhance gut motility if aspirates are more than 100mL. • Return the aspirate. • Flush the tube with 30mL water. Continue feeding. • Ensure the patient is nursed at 30-45 degrees head up as clinically indicated.
• 151 to 200mL on 2 or more consecutive occasions	• Return the aspirate. • Monitor closely for signs of abdominal distension, fullness, nausea, cramping and vomiting. • Liaise with MO and dietitian. • Consider medications to enhance gastric motility. • Ensure the patient is nursed at 30-45 degrees head up as clinically indicated.
• More than 300mL	• Return the aspirate • Flush the tube with 30mL water • Ensure the patient is nursed at 30-45 degrees head up as clinically indicated • Withhold the feed for 1 hour then re-check GRV If GRV is less than 300mL: Return the aspirate Flush the tube with 30mL water Deduct the aspirate amount from the scheduled feed If after 1 hour the aspirate is greater than 300mL: 1. Withhold the feed for 4 hours 2. Recheck GRV 3. Monitor closely for signs of abdominal distension, nausea and vomiting 4. Liaise with MO 5. Consider medications to enhance gastric motility. 6. Liaise with dietitian to discuss alternative feeding options e.g. reduced feeding rate or pump feeding.

10. Unblocking tubes

10.1 Procedure

- Liaise with shift coordinator /MO.
- Ensure there are no kinks in the tube.
- Refer to manufacturer's recommendations.
- Decant the water for flushing into the disposable cup.
- Attempt to administer 30-50mL of water ([refer to Feed Administration](#)) down the tube and leave for up to 30 mins. Repeat.
 - Alternatively, attempt to aspirate the introduced water immediately and repeat the procedure using gentle push and pull motions until the tube unblocks.
- Consider using warmed water.
- Avoid excessive force.
- If the tube remains blocked liaise with the MO or Senior/Specialist Nurse.
 - Consider the patient's fluid needs at this time.
 - Consider the use of a prescribed product to unblock the tube (e.g., Creon®, not suitable for medication blockages).
 - Refer to Specialist Nurse or duty gastroenterologist if available.
 - Consider replacing the tube if the above are unsuccessful.

11. Management of gastrostomy tubes

Refer to medical post-operative orders for management of the patient after insertion. Correct placement of the gastrostomy tubes are usually confirmed at insertion. Radiological confirmation may be necessary prior to use for some tubes.
Do not rotate pig-tail type radiologically placed gastrostomy tubes.

11.1 Post Insertion

- Ensure appropriate referrals are made e.g., to stoma nurse, dietitian.
- Assess the insertion site (stoma) by gently lifting the external bumper and observe for bleeding, signs of inflammation/infection, excoriation or leakage.
- Note the amount and type of leakage and volume of blood loss. If excessive leakage (greater than 50mL) is present, liaise with MO.
- Ensure the external tube length is marked and measured from the external bumper to the base of the feeding adaptor so that migration of the tube can be monitored.
- Cross-check the measurement with the measurement on the post-operative orders and liaise with senior staff/MO if there is a difference.
- Check and document the position of the external bumper. Ensure that it is approximately 1cm from the skin with no underlying dressing or compression to the wound.
- Liaise with senior staff/MO if migration of the external bumper occurs or it appears to compress the wound site.
- The patient should take nothing orally for 2 hours or via the tube for the first 4 hours.

- Bowel sounds should be checked for jejunostomy tube insertions. If bowel sounds are not present, liaise with MO and continue fasting the patient, including medications. If bowel sounds are present, flush the tube with sterile water at 4 hours and thereafter according to surgical care plan.
- Nurse 30-45 degrees head up for the water flushes and for 30 minutes afterwards as clinically indicated.
- Medications can be administered from 4 hours post insertion.
- Commence feeding as per the dietetic regime or as per MO orders.
- Flush the tube with 30mL of sterile water pre and post medication/feed.
- Dressings are not routinely required but may be necessary until post-procedure bleeding is stabilised.
- Provide the patient with relevant education and information leaflets.

12. Stoma Care

- During the initial post-procedure period, a sterile dressing is only required if there is post-procedure bleeding.
- Clean around the insertion site and under the external flange according to site, Sodium Chloride 0.9% is recommended in the post-procedure period and use of soap is not recommended until the insertion site has healed.
- Ensure the site dries thoroughly. Leave open to the air.
- If excessive stoma leakage is present, an absorbent foam dressing can be used.

12.1 Securement

- Between 2 and 4 T-fasteners or sutures may be inserted percutaneously to assist with the insertion and securing of radiologically inserted gastrostomy tubes.
- Observe the surrounding skin and dressing at clinically indicated intervals but at least at 24-48 hours after insertion and at 7 days after insertion. Refer to manufacturer's instructions.
- T-fasteners should be removed approximately 2 weeks after insertion with the following procedure:
 - Apply aseptic principles.
 - Clean around the T-fastener with antiseptic solution.
 - Lift the T-fastener away from the skin with forceps
 - Cut the thread at skin level with a stitch cutter or similar.
 - No dressing is required.
 - The remnant of thread left in the stomach will pass naturally.
- Patients discharged prior to removal of T-fasteners should receive information regarding their removal by health care providers in the community.
- Sutures should dissolve in 2-3 weeks. Suture locks should come away freely or may be cut. Liaise with senior/specialist nursing staff/MO.
- Gastrostomy tubes are not routinely secured to the abdomen externally. Agitated patients or those at risk of dislodging their tube may require the tube to be anchored.
- Use a drain securement device. Refer to manufacturer's information.
- Alternatively, an adhesive non-woven dressing can be used e.g., Fixomull®.

12.2 Daily Care

Do not rotate pig-tail type radiologically inserted gastrostomy (RIG) tubes

- Assess every shift for signs of leakage, inflammation, infection or pain at the stoma site.
- Measure the external tube length and bumper position and record in medical record.
- Patients with gastrostomy tubes can shower without soap. Ensure the area is dried well afterwards.
- PEG sites should be cleaned daily with mild soap and water. Allow to dry thoroughly and leave exposed.
- From 2 weeks after insertion, PEG tubes should be rotated 360 degrees. Once suture locks have detached or T-fasteners have been removed, RIG tubes without pigtails should be rotated daily.
- Ensure there is sufficient mobility with the tube to prevent unnecessary trauma.
- Ensure balloon tubes are checked weekly for balloon volume. Refer to Balloon Integrity Check section.
- Check the position of the external bumper. It should be approximately 1cm from the skin.
- Liaise with the MO as clinically indicated.
- Perform other specific cares:
- Perform mouth care every 2 to 4 hours as clinically indicated.
- Perform oro-pharyngeal suctioning as clinically indicated
- Patients should be weighed at least weekly or as clinically indicated.

13. Troubleshooting

13.1 Leakage around the tube

- Leakage of gastric/intestinal contents can cause chemical burns to surrounding skin.
- Leakage may occur when:
 - The internal bolster is incorrectly positioned: Check the external length of the tube. Gently pull the internal portion of the tube until resistance is felt then secure to prevent displacement. Check tube markings.
 - Buried flange: Ensure the external bumper is no more than 1cm from the skin. Try to rotate and advance the gastrostomy tube.
 - The balloon is deflated: Check the volume of water and compare to volume instilled. Refer to Balloon Integrity Check section
 - There is backflow of formula. Ensure the patient is positioned semi-upright as clinically indicated or liaise with the dietitian regarding slower rate of feeding or continuous feeding.

13.2 Pain during feeding/flushing through the gastrostomy tube

- May suggest mal positioning of the tube and/or leakage of gastric/intestinal contents into the peritoneal space.
- Notify the MO.

13.3 Balloon integrity check

This procedure is only to be performed on replacement tubes.

Some radiologically inserted (RIG) tubes and initial PEG tubes do not have balloons. Refer to manufacturer's information.

The maximum volume of the balloon must not exceed the manufacturer's instructions

Do not use air to inflate the balloon.

- Refer to manufacturer's instructions for the frequency and procedure for balloon integrity check.
- Check the documented volume instilled in the balloon at the time of insertion.
- Stabilise the tube and carefully remove fluid from the balloon with a syringe.
- Check the volume removed corresponds with the volume instilled at insertion.
- If the volumes correspond, re-instil the water into the balloon. If the volume is less than the original, instil additional fluid until the desired volume is reached.
- Liaise with MO/Senior Nurse if there are significant discrepancies between volume checks which may suggest balloon leakage.
- The tube will require replacement if the tube is not intact.

13.4 Gastrostomy tube dislodgement

Dislodged gastrostomy tubes must be attended to **immediately**.

- Liaise with MO to manage the patient's immediate need for hydration while awaiting tube replacement.
- If a gastrostomy tube appears to be dislodged but still in the gastrostomy tract, resecure it. Contact the radiology registrar to discuss.
- If a gastrostomy tube appears to be dislodged from the tract, a tube of the same size should be inserted immediately or at least within 1 hour to maintain the tract patency. Liaise with MO/Senior or Specialist Nurse.
- If a replacement gastrostomy tube is not available, temporarily insert a Foley urinary catheter of similar size and arrange for a planned gastrostomy tube re- insertion by an experienced staff member within 72 hours.
- Do not use a temporary tube until placement is confirmed.
- If a gastrostomy tube is not able to be replaced, liaise with duty gastroenterologist.
- Ensure adequate follow-up is in place by specialist medical/nursing staff for patients discharged from Emergency or short stay wards following replacement of dislodged gastrostomy tubes.

13.5 Gastronomy tube removal/placement

Note: Assess for flange type. Some gastrostomy tubes can only be removed endoscopically or by trained staff. Liaise with senior/specialist nurse or MO. Refer to manufacturer's instructions.

- Medical and Nursing Staff performing gastrostomy tube removal and changes must have appropriate training.
- A Gastrostomy tube is not routinely changed and may remain in-tact for up to 2 years if viable.
 - Tubes should not be changed until the stoma tract is completely healed or at least 6 weeks after placement.
- Tubes can be replaced under fluoroscopy by request of MO.

A replacement tube must be available on the ward which is of a similar size to the gastrostomy tube inserted.

- Normally 18-20 French gauge or Foley 16-18 French gauge.
- Consider need to withhold treatment for patients who are anticoagulated.
- If the patient is deceased, the gastrostomy tube should remain in-situ unless removal is ordered by MO.

13.6 Equipment

- Gastrostomy replacement tube
- Antiseptic solution e.g., chlorhexidine 1%
- Oral / enteral syringes 20mL and 50-60mL
- Lignocaine gel and water-based lubricant
- Dressing pack
- pH indicator strips
- Sterile water
- Waterproof absorbent sheet

13.7 Pre-procedure

- Fast the patient for 6 hours prior to the procedure.
- Cease anti-coagulant therapy at the clinically appropriate time.
- Explain the procedure to the patient and gain appropriate consent.
- Position the patient in a comfortable reclined or supine position as clinically indicated.
- Perform an aseptic non-touch technique, refer AKG Aseptic Technique Procedure.
- Open the sterile dressing pack and place equipment on the sterile field.
- Prepare the new tube for insertion:
 - Remove from the packaging
 - Check the balloon by inflating/deflating
 - Lubricate the end of the tube
 - Ensure the external flange is withdrawn up into the tube.

13.8 Removal of the gastrostomy

13.8.1 Tube with bumper:

- Note the position of the external flange on the old tube prior to removal.

- Open the end of the gastrostomy tube.
- Ensure the tube is mobile in the tract with adequate movement for manipulation.
- Clean the stoma with antiseptic solution and dry
- Hold the gauze dressing over the opening to absorb any fluid.
- Using one hand, grip the tube. Place the other hand on the abdomen surrounding the stoma.
- Using counter traction, gently but firmly pull the existing tube out.
- Liaise with the MO/Senior Nurse if the tube is difficult to remove.

13.8.2 Tube with balloon:

- Open the end of the gastrostomy tube.
- Ensure the tube is mobile in the tract with adequate movement for manipulation.
- Clean the stoma with antiseptic solution and dry.
- Deflate the balloon fully.
- Hold the gauze dressing over the opening to absorb any fluid.
- Using one hand, grip the tube. Place the other hand on the abdomen surrounding the stoma.
- Using counter traction, gently but firmly pull the existing tube out.
- Liaise with the MO/Senior Nurse if the tube is difficult to remove.

13.8.3 If the gastrostomy is no longer clinically indicated:

- Clean the stoma again with antiseptic solution and dry
- Apply a dry dressing to the stoma and dress daily
- Observe for ooze or excessive exudate
- Liaise with MO/Senior Nurse as appropriate
- Provide dressings and education for discharge as appropriate.

13.9 Replacement tube Insertion

- Insert the new lubricated tube into the stoma approximately 5-6cm beyond the previous flange position to avoid obstructing the gastric outlet or the stoma.
- Inflate the balloon with sterile water. Refer to manufacturer's instructions.
- Gently tug the tube to ensure the balloon is inflated and secure.
- Push the external flange down the tube towards the abdomen.
- Leave a 1cm gap between the bumper and the skin.
- Aspirate the gastrostomy tube and check pH to confirm correct tube placement prior to use.
- If unsure of tube placement, do not use. Liaise with MO.
- A follow-up pegogram may be indicated.
- Monitor the site daily for signs of bleeding, infection or leakage

14. Feed administration

- The nasally or orally inserted gastric tubes must be checked to confirm correct placement in the stomach prior to each feed. Position is confirmed with gastric aspirate pH testing and/or x-ray. Refer to Checking Tube Placement.

- Check the gastrostomy tube length for migration prior to each feed, water flush or medication administration.
- If unsure of placement do not use the enteral tube. Notify Senior Nurse/ MO.
- For healthy, immune competent and orally fed patients tap or bottled water may be adequate for:
 - Flushing the tube before and after each change of feed, or bolus feed.
 - Before and after tube aspiration.
 - When administering medications via the tube.

Exceptions being:

- Immunocompromised patients and those being fed post pylorically (including jejunostomy and gastro-jejunostomy tubes) – freshly opened bottled **sterile** water should be used.
- Breaking the feed circuit to top up a bag. If feeds are to be diluted, sterile water is recommended.
- All duodenal and jejunum feeds must be administered continuously by pump.
- Check for patient food allergies and if present, check formula for allergens contained prior to administration, refer to [Appendix 4 AKG Enteral Feed Allergen Matrix](#).
- Ensure the formula is at room temperature.
- Refer to manufacturer's information for the correct use of feeding delivery systems.
- Observe the patient during feeding for:
 - Signs of respiratory difficulty or vomiting/regurgitation and report to MO/senior nursing staff immediately or escalate care e.g., MET call.
 - Tolerance during feeding including abdominal distension, discomfort or pain, or diarrhoea.
- Do not use feeds if past expiry date.

14.1 Feeding procedures

- Explain the procedure to the patient/carer(s)/NOK.
- Wipe the feeding solution packaging with antiseptic wipe prior to opening. Allow to air dry.
- Position the patient with their head raised 30-45 degrees as clinically indicated. Maintain this position for at least 30-60 minutes after the feed.

14.2 Equipment required:

- Feeding regimen documented by dietitian, AKMR 60.3 Enteral Feeding Regime
- Gravity feeding set (if feeding by delivery set)
- Enteral feeding pump
- 50mL enteral syringe
- Disposable cup
- Water
- Prescribed feeding formula
- Clean receptacle for the aspirate
- Pump feeding delivery set (as required)

- Waterproof protective sheeting
- pH testing strips

14.3 Feeding via enteral syringe

- Check nasally/orally inserted gastric tube placement if feeding through this.
- Measure gastric residual volume for nasogastric tubes placed in the stomach.
- The syringe can be attached to the feeding tube with the barrel removed and the tube clamped/kinked.
- Flush the enteral tube with water.
- Pour the feeding solution into the barrel of the syringe.
- Release the cap, clamp or kink to allow the feed to flow through the tube using gravity.
- Cap, clamp or kink the tube and repeat the procedure of administering the feeding solution until the prescribed amount has been delivered.
- Flush the tube again with at least 30mL of water or volume as per AKMR60.3 Enteral Feeding Regime.
- Clamp or kink the tube.
- Replace the cap/spigot.

14.4 Feeding via gravity set

- Check nasally/orally inserted gastric tube placement if feeding through this.
- Measure gastric residual volume for nasogastric tubes placed in the stomach.
- Prepare the delivery set and formula delivery device. Refer to manufacturer's instructions.
- Ensure no part of the feeding set container that comes into contact with the feeding formula is touched.
- Fill the feeding bag with the prescribed bolus volume. Label the bag with date and feed type.
- If feeding bottles are being used. Ensure details are correct on the bottle.
- Flush the enteral tube with at least 30ml water or volume as per AKMR60.3 Enteral Feeding Regime.
- Cap, clamp or kink the tube and attach the delivery set. Ensure it is secure.
- Administer the feed regulated by the gravity feeding set.
- Following the feed, flush the tube again with water.
- Disconnect the delivery set and re-cap/spigot the tube.

14.5 Continuous feeding via pump set

- Check nasally/orally inserted gastric tube placement if feeding through this.
- Measure gastric residual volume for nasogastric tubes placed in the stomach.
- Flush the enteral tube with 30mL of water.
- Insert the pump feeding set into the pump according to manufacturer's instructions.
- Connect the pump delivery set line to the feeding tube. Ensure it is secure.
- Ensure the line is primed prior to connection to the patient
- Administer feed according to the regimen prescribed.
- Flush gastrostomy tubes situated in the jejunum every 2-4 hours during continuous feeding.

- Nasally and orally inserted gastric tubes should be aspirated and flushed every 4 hours to check GRV, refer to [Aspiration and Management of Gastric Residual Volume](#).
 - However, the effect of these assessments on daily nutritional intake should be considered as clinically indicated. Liaise with the team.
- Recommence the feeding regimen as ordered.
- Notify the dietitian of any significant stop in feeding.
- Consider pH testing of gastric aspirate from an OG/NGT following prescribed breaks and prior to feeding set change/recommencement of feeding. Wait until at least 1 hour after the feed has stopped.
- Feeding sets must be changed a minimum of once each day.
- Ready to hang (RTH) feeds are closed systems and can be hung at room temperature for up to 24 hours. Change the feeding set each time a new pre-filled RTH feed is used.
- For other feeds e.g., bottle, can or tetra Pak place a maximum of 4 hours of feed in the feeding bag and only top up the feeding bag set when it is almost empty. Open feeds can be sealed and refrigerated for up to 24 hours.
- Label the feeding bag with feed type, date, time and rate. Re-label the feeding bag if a rate change is prescribed.
- OG/NGT placement is checked at a minimum every 24 hours for patients on continuous feeds. Also check prior to medication administration and re- commencement of feeds.

15. Blended tube feeds

- Strictly for those patients who are already established on a blended tube feed (BTF) regimen at home.
 - BTF is not standard practice at AKG however if a patient is already established on BTF at home, then support and advice will be provided to ensure the patient is safely meeting their nutritional requirements.
 - Risks of BTF can include, but are not limited to, feeding tube occlusion, infection, bacterial overgrowth and nutritional deficiencies.
 - Administration of BTF by nursing staff will require approval from the Nurse Unit Manager.
- Patients admitted across AKG on an established BTF regime requesting to continue BTF within the hospital setting must be referred to the ward dietitian.
 - The dietitian will assess the patient's nutritional status and adequacy of recipe/s being used.
 - The decision for the patient to continue with BTF whilst admitted at AKG is a joint decision between the treating team and the dietitian. The decision is also dependent on the patient's clinical condition.

15.1 Feeding procedures

- The patient/parent(s)/carer(s) are required to:
 - Take full responsibility for the feed, including preparation (off-site), transport, storage, administration and cleaning of equipment throughout the admission.
 - Sign an agreement, acknowledging the potential risk associated with BTF administration and take full responsibility. Refer to [Appendix 6 Blended Tube Feed – Intent of Use Statement in an Acute Care Setting](#).

- Due to the variation in how patient/parent(s)/carer(s) use the BTF, it is advisable that nursing staff observe the administration process by the patient/parent(s)/carer(s) in the first instance to ensure good practice.
- The dietitian will clearly document:
 - The BTF regimen, including administration parameters as agreed upon with the patient/parent(s)/carer(s) and the treating team.
 - Alternative regimen using nutritionally complete sterile enteral nutrition formula, in the event there is inadequate supply of BTF or lack of appropriate carer(s) to administer the BT, or the decision to use the BTF is revoked.

15.2 Exclusion Criteria

- Patients not already on established BTF at home.
- Patients who are:
 - Immunocompromised
 - At a high risk of infection
 - Malnourished
 - Requiring fluid restriction
- Patients who require continuous/intermittent feeds or inability to tolerate bolus feeding.

15.3 Practical Recommendations

- All fresh food items should be transported in a heat pack to ensure foods are kept at correct temperatures (over 60°C for hot food) to avoid food related illnesses.
- Reheating facilities are not available for use on the ward.
- All BTF should be prepared as close to administration time as possible and should not remain at room temperature for longer than 2 hours (this includes time for administration of feed).
- A slow plunge method, using a 60ml enteral syringe, is the method of choice for the administration of BTF.
- It should be noted feeding equipment manufacturers/suppliers have advised against the administration of BTF using their equipment.
- Commercially available BTF (e.g. Wholesome Blends)
- Patient/carers may choose to use commercial real food options for enteral feeding instead of homemade BTF.
- Should the nursing staff be required to administer the commercial real food options for enteral feeding, they will be required to check the use-by date.

16. Medication administration

- Use only oral and enteral syringes intended and labelled for the administration of oral liquid medications.
- For fine bore intestinal tubes (naso-duodenal/naso-jejunal, jejunostomy), liaise with the Pharmacist for appropriate medication preparation to prevent clogging e.g. liquid form preferred.
- If unsure of placement do not use the enteral tube for medication administration. Notify Senior Nurse/ MO.

- **Do not** add medications to feeding formula. Liaise with the Pharmacist.

- Consider drug interactions and drug/nutrient interactions.
- Consider alternative routes for medications and use liquid preparations where possible.
- Some medications are not designed to be crushed, Liaise with Pharmacist/MO and refer to Don't Rush to Crush via "Crush" tab on [MIMSONline](#).
 - Buccal or sublingual medications
 - Enteric coated medications
 - Slow release medications
 - Hormone preparations
 - Cytotoxic medications
 - Proton pump inhibitors
- Liaise Pharmacist for the following:
 - Medications that interact with feeding formula.
 - Interruption of feeding for medications that are administered via the enteral tube.
 - Medications to be taken on an empty stomach.
 - Enteral tube compatible formulations of medications.
- For patients with a gastric tube on straight drainage:
 - Liaise with the Pharmacist, Dietitian and/or MO regarding cessation of feed prior to specific medications.
 - Clamp/spigot the tube for at least 30 minutes after administration of the drug.
- Consider sugar and sodium content in some liquids/soluble forms.
- Consider the amount of fluid given to patients on fluid restriction.

16.1 Equipment required:

- Inpatient medication chart
- pH test strips
- 50mL enteral syringe
- Clean pill crusher or mortar and pestle
- Disposable cup(s)
- Water – refer to feed administration
- Disposable receptacle
- Plastic-backed absorbent sheet
- Sterile scissors / blade

16.2 Procedure

- Explain the procedure to the patient and gain appropriate consent
- Position the patient comfortably with their head raised 30-45 degrees as clinically indicated.
- Pour water into the disposable cup(s).
- Check nasally/orally inserted gastric tube placement if administering medication through this.
- Prepare the prescribed medication for administration according to instructions in Don't Rush to Crush via "Crush" tab on MIMS Online.

- Flush the enteral tube with 15-30mL of water.
- Draw up the prepared diluted medication into the enteral syringe, promote mixing.
- Release the cap, clamp or kink and administer the medication by gravity or gentle push.
- Re-place the cap, clamp or kink the enteral tube and disconnect the enteral syringe.
- If more than one medication is prescribed to be administered at the same time:
 - Prepare each medication individually.
 - Flush between each medication with 15mL of water.
- Perform the post medication administration water flush to clear enteral tube.
- Clarify if fasting is required following medication administration before re-starting the feed.

16.3 Gastric decompression

- Can be monitored by:
 - NGT on straight drainage, including aspirates every 4 hours as clinically indicated.
 - NGT capped/spigoted, including aspirates every 4 hours as clinically indicated.
- The drainage bag should be changed every 24 hours or as required when there is large drainage.

17. Documentation

Minimum documentation standards required:

17.1 Orally and nasally inserted tubes

17.1.1 At insertion:

- Date and time of insertion
- Reason for insertion
- Type of tube
- Gauge/size of tube
- External length of tube
- Initial confirmation of tube position via x-ray

17.2 After checking placement/aspiration/pH testing/drainage

- Gastric residual volume on fluid balance chart, amount returned/discarded
- Gastric drainage volume
- pH levels
- Any tube manipulation

17.3 After medication administration

- Complete medication and fluid balance charts

17.4 After removal

- Document the reason for removal and date and time of removal tube

17.5 Gastrostomy tubes

- Brand, size and type of tube inserted
- Post procedure vital signs/observations
- External length of tube and position of external bumper
- Signs of leakage, skin integrity, pain and infection at the stoma site
- Amount of water balloon if balloon type and balloon volume checks, including volume removed and re-instilled

17.6 Enteral Feeding

- Enteral feeding regime on AKMR60.3 Enteral Feeding Regime
- Name of feed and amount administered
- Water administered
- Frequency and strength of feeds
- Pump feeding set changes
- Complications with feeding

18. Compliance monitoring

Compliance against this procedure will be evaluated by clinical incidents registered with DATIX Clinical Incident Management System.

19. Policy context

The following documents are required to give effect to this procedure:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- HDWA [Clinical Handover Policy \(MP 0095\)](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching Policy \(EMHS: 21\)](#)

20. Supporting information

The following documents support the implementation of this procedure:

- [Aseptic Technique Procedure \(AKG-PRO-0045-14\)](#)
- [Medication Policy \(AKG-POL-0253-13\)](#)
- [Medication Management Procedure \(AKG-PRO-0255-18\)](#)
- [Nutrition and Hydration Policy \(AKG-POL-0603/23\)](#)

21. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.1 Integrating clinical governance.
- Standard 5 – Action 5.27 Nutrition and hydration.
- Standard 5 – Action 5.28 The workforce uses the systems for preparation and distribution of food and fluids.

22. References

1. Dietitians Association of Australia, June 2018. Enteral feeding manual for adults in health care facilities. Dietitians Association of Australia, Canberra.
2. Malnutrition Advisory Group, British Association of Parenteral and Enteral Nutrition. Malnutrition Universal Screening Tool. The Malnutrition Universal Screening Tool (MUST) is reproduced here with the kind permission of BAPEN October 2008. www.bapen.org.uk
3. Nguyen NQ et al. Prokinetic therapy for feed intolerance in critical illness: One drug or two? Crit Care Med 2007; 35(11): 2561-2567.
4. Reeves A et al. Multidisciplinary evaluation of a critical care enteral feeding algorithm. Nutr Diet 2012; 69: 242-249.
5. Singer P et al. ESPEN guideline on clinical nutrition in the intensive care unit. Clinical Nut 2019; 38: 48-79

23. Glossary

The following definitions are relevant to this procedure:

Term	Definition
Enteral Nutrition	Enteral nutrition support refers to the delivery of a nutritionally complete feed directly into the stomach or small intestine using a narrow tube
Gastric Tube	Enteral tube resting in the stomach
Intestinal Tube	Enteral tube resting in the duodenum or jejunum
Nasogastric Tube (NGT)	Nasally inserted gastric tube
Nasojejunal Tube (NJT)	Nasally inserted intestinal tube inserted into the jejunum
Gastrostomy	Enteral tube which has been inserted directly through the abdominal wall into the stomach, can include Percutaneous Endoscopic Gastrostomy (PEG), Radiologically Inserted Gastrostomy (RIG)
Blended Tube Feed (BTF)	Home made blended enteral formula, alternative descriptions include blenderised feeds, blended food, liquidised tube diet and puree tube foods.

24. Document contact

Enquiries relating to this procedure may be directed to:

Title: Coordinator Dietetics and Nutrition
 Department: Hospital Wide
 Email: jane.priest@health.wa.gov.au

25. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

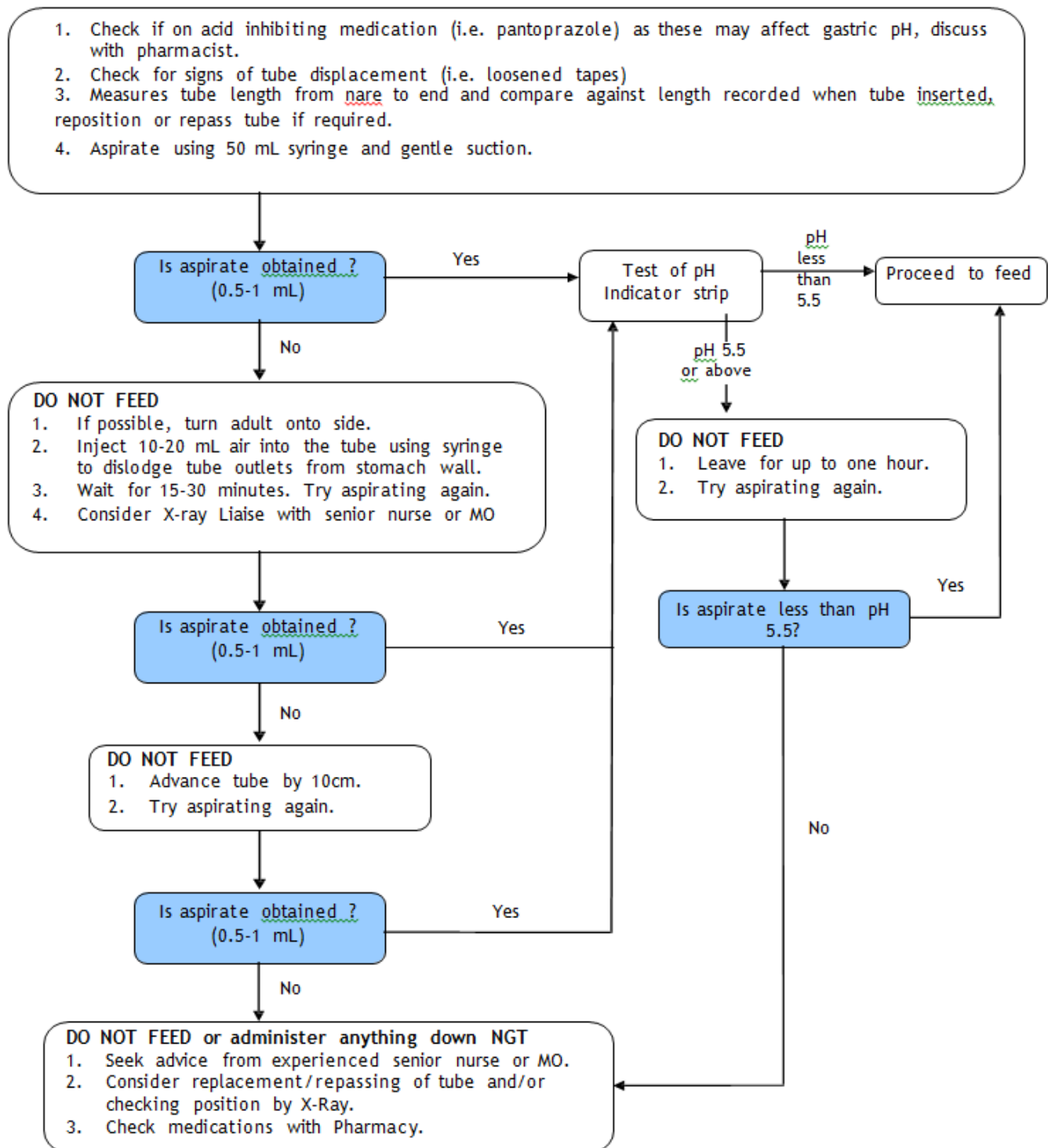
Version	Effective from	Effective to	Amendment(s)
4	18/07/2024	18/07/2027	

26. Authorisation

This procedure has been authorised and issued by:

Authorised by	Danielle Kilmurray – Director Allied Health and Ambulatory Care
Authorisation date	18/07/2024
Published date	18/07/2024

Appendix 1: Algorithm for Confirming the Correct Position of the OGT/NGT in Adults



Appendix 2: Hospital Wide Adult After-hours Enteral Feeding Regime

Use Isosource 1.5 feed for all enteral starter regimes

Day	Hours	Feed Formula	Strength	Rate	Total Volume (ml)	Water Flushes (ml)	Total Fluid (ml) per day
1	50ml Water flush prior to commencing feeds						
1	0600-2200	Isosource 1.5	Full	20ml/hr	320	50 x 5	757
2	0600-2200	Isosource 1.5	Full	33ml/hr	528	50 x 5	778
3	0600-2200	Isosource 1.5	Full	46ml/hr	736	50 x 5	986

Continue with above regime until reviewed by Dietitian

- 50ml Water flushes every 4 hours (0600,1000,1400,1800,2200), prior to commencing feeds and post feeds. Stop feeds, provide water flush via syringe then recommence feeds.
- Flush tubes with at least 30ml of water:
 - Before, between and after medication administration and before and after checking gastric residuals.

Re-Feeding Syndrome Protocol

*****If a patient has had nil intake or been fasted for >5 days**

Please feed using the feeding schedule below

Day	Hours	Feed Formula	Strength	Rate	Total Volume (ml)	Water Flushes (ml)	Total Fluid (ml)
1	50ml Water flush prior to commencing feeds						
1	0600-2200	Isosource 1.5	Full	14ml/hr	224	50 x 5	474
2	0600-2200	Isosource 1.5	Full	20ml/hr	320	50 x 5	757
3	0600-2200	Isosource 1.5	Full	26ml/hr	416	50 x 5	666

Continue with above regime until reviewed by Dietitian

- **Supplementation:** Oral thiamine 200-300mg daily
- B Group vitamin, complex tablets OR full dose daily IV Vitamin B Multivitamin/trace element supplement once daily
- 50ml Water flushes every 4 hours (0600,1000,1400,1800,2200), prior to commencing feeds and post feeds. Stop feeds, provide water flush via syringe then recommence feeds.
- Flush tubes with at least 30ml water:
 - Before, between and after medication administration and before and after checking gastric residuals.

Refer patient to the ward Dietitian on Extension: 2536

Appendix 3: Intensive Care Unit (ICU) After-hours Enteral Feeding Procedure

1. Confirm with the Duty Medical Officer if the patient is appropriate for enteral feeding.
2. Check sodium, potassium, magnesium and phosphate prior to commencing feeding regime and supplement as necessary and continue to monitor daily.
3. Record patient's height and weight at time of admission or as soon as clinical condition permits.
4. Determine patient's IBW, based on height or estimated height from ulna length (refer to table below), record IBW and ulna length on *AKMR 60.3 Enteral Feeding Regime*.
5. If current actual weight is less than calculated IBW, use actual weight to determine feeding requirements.
6. If the patient has not eaten or been fasted for 5 days or more, they may be at risk of refeeding syndrome. If patient is at risk of refeeding syndrome thiamine should be supplemented and phosphate, calcium, magnesium and potassium levels are required to be checked prior to commencing enteral feeding and replaced if necessary. Continue to monitor daily until corrected.
7. Prior to using the feeding tube document, the date and time of placement, placement confirmation from medical team, aspirate pH and external length of NGT.
8. Check for food allergies, select appropriate enteral feed and commence enteral feeding as per ICU After-Hours Enteral Feeding Protocol.
9. Document the choice of the enteral feed formula and target feeding rate in the patient's medical record and on *AKMR 60.3 Enteral Feeding Regime*.
10. Contact the medical officer regarding additional fluids to ensure fluid requirements are met.
11. Contact the dietitian to review the patient as early as possible (earliest working day).

Additional Considerations - All starter regimes are 24-hour continuous regimes.

How to estimate height from ulna length

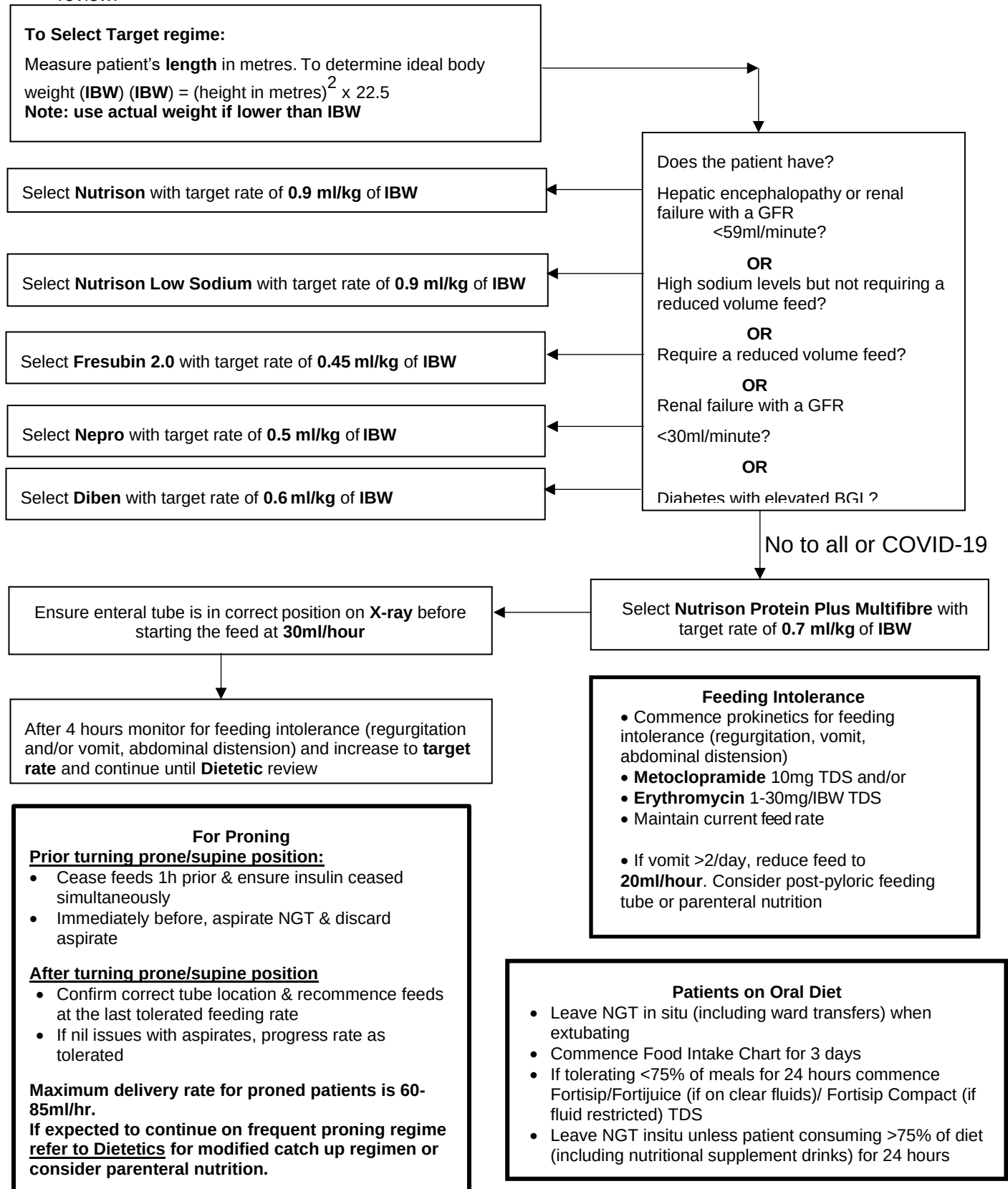
Measure between the point of the elbow (olecranon process) and the midpoint of the prominent bone of the wrist (styloid process) left side if possible.

Height (m)	men (<65 years)	1.94	1.93	1.91	1.89	1.87	1.85	1.84	1.82	1.80	1.78	1.76	1.75	1.73	1.71
	men (≥65 years)	1.87	1.86	1.84	1.82	1.81	1.79	1.78	1.76	1.75	1.73	1.71	1.70	1.68	1.67
	Ulna length (cm)	32.0	31.5	31.0	30.5	30.0	29.5	29.0	28.5	28.0	27.5	27.0	26.5	26.0	25.5
Height (m)	Women (<65 years)	1.84	1.83	1.81	1.80	1.79	1.77	1.76	1.75	1.73	1.72	1.70	1.69	1.68	1.66
	Women (≥65 years)	1.84	1.83	1.81	1.79	1.78	1.76	1.75	1.73	1.71	1.70	1.68	1.66	1.65	1.63
Height (m)	men (<65 years)	1.69	1.67	1.66	1.64	1.62	1.60	1.58	1.57	1.55	1.53	1.51	1.49	1.48	1.46
	men (≥65 years)	1.65	1.63	1.62	1.60	1.59	1.57	1.56	1.54	1.52	1.51	1.49	1.48	1.46	1.45
	Ulna length (cm)	25.0	24.5	24.0	23.5	23.0	22.5	22.0	21.5	21.0	20.5	20.0	19.5	19.0	18.5
Height (m)	Women (<65 years)	1.65	1.63	1.62	1.61	1.59	1.58	1.56	1.55	1.54	1.52	1.51	1.50	1.48	1.47
	Women (≥65 years)	1.61	1.60	1.58	1.56	1.55	1.53	1.52	1.50	1.48	1.47	1.45	1.44	1.42	1.40

Accessed from www.bapen.org.uk

ICU After-Hours Enteral Feeding Protocol

This flow chart can be used to initiate early enteral feeding until the dietitian is available to review the patient. Enteral feeding to be commenced cautiously in underweight, severely malnourished or starved individuals as they may be at risk of 'refeeding syndrome'. Please contact the dietitian on page 226 to review.





Appendix 4: AKG Enteral Feed Allergen Matrix

C indicates menu item **CONTAINS** the allergen as an ingredient.

PAL stands for Precautionary Allergen Labelling statement.

											Tree nuts								Gluten containing cereals					
Product / item	Date Assessed	Milk (dairy)	Egg	Fish	Crustacean	Mollusc	Sesame	Lupin	Peanut	Soy	Almond	Brazil nut	Cashew	Hazelnut	Macadamia	Pecan	Pine nut	Pistachio	Walnut	Wheat	Gluten	Barley	Oats	Rye
Fortisip	05/24	C								C														
Fresubin 2	05/24	C								C														
Fresubin 2 fibre	05/24	C								C														
Nepro HP	05/24	C								C														
Isosource 1.5	05/24	C								C														
Diben	05/24	C		C						C														

Before referencing this procedural document please ensure that you have the latest version from the [Policy Library](#) information hub page.

Enteral Feeding and Enteral Tubes Management – Adult Procedure

											Tree nuts								Gluten containing cereals					
Product / item	Date Assessed	Milk (dairy)	Egg	Fish	Crustacean	Mollusc	Sesame	Lupin	Peanut	Soy	Almond	Brazil nut	Cashew	Hazelnut	Macadamia	Pecan	Pine nut	Pistachio	Walnut	Wheat	Gluten	Barley	Oats	Rye
Nutrison	05/24	C		C						C														
Nutrison Low Sodium	05/24	C								C														
Nutrison Protein Plus Multifibre	05/24	C		C						C														
Isosource 2	05/24	C								C														
Beneprotein	05/24	C								C														



Appendix 5: ICU Only – Catch up feeding procedure

Step 1: When restarting feeds recalculate the target rate for the remaining hours in the 24-hour period, document on AKMR Catch Up Enteral Feeding Calculation Form 60.3.2.

Table 1: Formula to determine catch up feed rate

24-hour Goal Volume:	
	Minus
Volume Already Received Today:	
	=
Volume Remaining for Today:	
	Divided by
Total Hours Remaining Today:	
	=
Rate Required to Provide Goal Volume:	

Example: Patient's 24-hour goal volume is 1200ml, feed rate over 24 hours is 50ml/h.

The patient is fed from 0700-1300 hours at 50ml/h, 300ml feed already received. Feeds are stopped from 1300-1700 hours, there are now 14 hours remaining to reach the goal volume by 0700 hours

24-hour Goal Volume:	1200
	Minus
Volume Already Received Today:	300
	=
Volume Remaining for Today:	900
	Divided by
Total Hours Remaining Today:	14
	=
Rate Required to Provide Goal Volume:	64

Key alert: Do not increase feed rate above 150ml/h.

Step 2

At the end of the 24-hour period return enteral feeding rate to daily target rate

Before referencing this procedural document please ensure that you have the latest version from the [Policy Library](#) information hub page.

Appendix 6: Blended Tube Feed – Intent of use statement in an acute care setting



Government of Western Australia
East Metropolitan Health Service

Armada Kalamunda Group

Blended Tube Feed – Intent of Use Statement

The decision was made with _____ [name patient/parent(s)/carer(s)] and the multidisciplinary team involved in their care _____ (Consultant)
_____ (Dietitian)
that _____ (Tube fed individual's name) will be fed using a 'Blended Tube Feed' via their Enteral Feeding Device.

I agree that I [name(patient/parent(s)/carer(s))] _____ will be administering the blended tube feed via a syringe through the Enteral Feeding Device. I understand that I am accountable for maintaining optimal food safety and hygiene practice. I acknowledge the potential risks associated with the administration of Blended Tube Feed and I take full responsibility.

Patient/ Parent(s)/ Carer(s) signature: _____

Patient/ Parent(s) / Carer(s) first and last name: _____

Consultant Signature: _____

Consultant first and last name: _____

Appendix 7: Potential problems associated with Enteral Tubes

• Nasoenteric/Oroenteric		• Gastrostomy/Jejunostomy Tubes	
• Nasogastric • Oro-gastric • Naso-duodenal • Naso-jejunal	• Incorrect positioning • Epistaxis • Sinusitis • Dislodgement of the tube • Aspiration pneumonia • Nasal Trauma • Tracheal perforation • Pneumothorax • Skin breakdown	• Percutaneous Endoscopic Gastrostomy (PEG) • Percutaneous Gastrostomy (radiologically inserted) (RIG) • Percutaneous Endoscopic Jejunostomy (PEJ)	• Wound infection • Bleeding, gastric/intestinal leakage • Inadvertent removal • Liver or bowel perforation • Intra-peritoneal air or bleeding • Ileus • Gastric outlet obstruction, ulceration • Colo- cutaneous fistula • Necrotizing fasciitis

Appendix 8: Potential problems associated with enteral feeding

• Problem	• Cause	• Considerations
Aspiration	Reflux or vomiting	Reduce feed rate or slow progression to target rate, avoid large volume bolus feeding
	Sub-optimal patient positioning	Ensure upright position (more than 30 degrees) where possible during and after feeding for at least 30 minutes
	Physical agitation e.g. while transferring	Stop feeding at least 30 minutes prior to mobilisation or vigorous activity
	Tube dislodgment	Check tube position regularly
	Medications e.g. anti-psychotics, anti-cholinergics	Review medications
	Other e.g. neurological disorders, altered consciousness level	Identify and address risk factors
Blocked feeding tube	Irregular tube flushing Concentrated or fibre-enhanced formula	- Flush tube before and after each feed or every 4 hours during continuous feeds - Flush before and after checking aspirates
	Incorrect administration of medications	Refer to 3.10 Medication Administration
	Feeding tube deterioration or kinking/twisting	Confirm tube positioning with x-ray as required Re-position or replace the NGT, liaise with MO
Constipation	Inadequate fluid	Liaise with the team. Review requirements and monitor intake and output. Consider extra flushes
	Inadequate fibre	Consider fibre enriched formula or fibre supplementation
	Disruption of usual diurnal cycle of food and activity, toileting	Discuss with the team, optimise mobility and attention to privacy
	Medications e.g. analgesics	Review medications, consider bowel management plan with laxatives

	Gastro-intestinal obstruction	Cease feeding and refer to medical team if obstruction is suspected
Dehydration	- Inadequate fluid intake - Excessive fluid loss	- Review fluid requirements and monitor fluid balance - Consider extra/larger flushes or intravenous fluids - Consider changing to less concentrated formula
Delayed Gastric Emptying	- Illness - Post-surgical stress response	- Assess tolerance with aspirate volume, abdominal distension, discomfort and nausea/fullness - Consider post-pyloric feeding or pro-kinetics
	Hyperglycaemia	Monitor blood glucose levels
	Medications	Review medications
	Gastro-intestinal obstruction	Refer to team and cease feeding if obstruction is suspected
Diarrhea	Antibiotics	- Discuss with MO - Consider probiotic supplement
	Medications	Review medications, withhold laxatives
	Fibre content of formula	- A fibre free feed or lower fibre content feed may be indicated (if previously on fibre feed) - Inclusion of fibre in feed may resolve diarrhea (if previously on no fibre feed)
	Bolus feeding or rapid administration	Use continuous administration or slower rate over a longer duration
	Feed administered too cold	Let formula stand at room temperature for 30 minutes prior to use
	Hyperosmolar formula	Some patients may be sensitive, check medication osmolality, consider iso-osmolar feed
	Malabsorption	Investigate the need for pancreatic enzyme replacement therapy
	Bacterial contamination	Adhere to hygiene precautions and hang time, avoid adding to the feed

	• Overflow Diarrhea (constipation)	• Check for impaction, confirm as clinically indicated with rectal exam/x-ray
Fluid Overload	• Excessive fluid intake • Compromised renal or cardiac function	• Review all fluid volume received including: - oral and intravenous fluids, medications and flushes, liaise with MO and titrate accordingly • Consider smaller flushes • Consider changing to a more concentrated formula • Document regular weight measurements, liaise with the team
Nausea and Vomiting	• Fast delivery rate/overfeeding/large volume	• Reduce the feed rate, reduce the amount of each bolus, consider continuous feeds, check nutritional requirements to prevent overfeeding
	• Delayed gastric emptying	• Consider post-pyloric feeding or pro-kinetic medication, liaise with MO
	• Sub-optimal patient feeding position	• Ensure upright position (more than 30 degrees) where possible during and after feeding for at least 30 minutes
	• Tube dislodgement	• Check tube position regularly
	• Medications e.g. opioids	• Review medications
	• Feeding formula too cold or concentrated	• Ensure feed is administered at room temperature, consider changing to isotonic formula
	• Physical agitation e.g. while transferring	• Stop feeding at least 30 minutes prior to mobilisation or vigorous activity
Drug nutrient interactions	• Examples of oral/enteral drug nutrient interactions may include but are not limited to, refer to MIMSONline for further information:	
	• Phenytoin • Moxifloxacin • Levothyroxine • Ciprofloxacin • Rifampicin • Voriconazole	• Discuss most appropriate route with MO, Pharmacist and Dietitian • Viscous medications should be diluted well • Stop feeds 2 hours before and resume 2 hours after administration • Alter feeding regimen to allow for breaks
	• Mono-amine oxidase inhibitors	• Discuss with MO, Pharmacist and Dietitian

	Warfarin	- Discuss with MO, Pharmacist and Dietitian Feeding formula components may reduce drug effectiveness - Dose requirement may change, adjust according to INR - Ensure to flush well pre and post administration - It may help to stop feed 1-2 hours before and after administration
Ileus/Reduced Gut Motility	Post-operative (Primary)	- Discuss with MO and - Should be able to resume enteral nutrition within 24 hours of trauma/surgery - Consider post-pyloric feeding if delayed gastric emptying
	Paralytic (Secondary)	- Discuss with MO and Dietitian to potentially cease enteral feeds - Consider TPN if enteral feeding unable to meet full nutritional requirements - Consider post-pyloric feeding if delayed gastric emptying - Monitor the patient for abdominal distension, pain, reflux, belching, nausea or increased aspirates
Abnormal biochemistry	Hypoalbuminaemia	- Check feeding is adequate for needs - Check for infection/inflammation - Observe trends rather than single results
	Hypercalcaemia	- Discuss with MO and Dietitian - Check fluid intake is adequate - Review supplementation especially if renal function is compromised
	Hyperglycaemia	- Discuss with MO, and Dietitian - Review the need for insulin or oral hypoglycaemic medications rather than altering nutritional intake - Consider low GI feed
	Hyperkalaemia	- Discuss with MO and Dietitian. Consider a low potassium formula - Review medications that cause retention. In rare circumstances consider reducing the feed rate where the gut is poorly perfused

	Hypokalaemia	- Discuss with MO and Dietitian. Review medications (diuretics) - Consider supplementation - Check fluid intake is not excessive - Manage vomiting, Diarrhoea and inadequate nutritional intake - Manage refeeding syndrome
	Hypernatraemia	- Discuss with MO and Dietitian. Review medications for sodium - Consider larger/more flushes
	Hyponatraemia	- Discuss with MO and Dietitian. Review fluid intake - Consider supplementation
	Hyperphosphataemia	- Discuss with MO and Dietitian - Review if excessive vitamin D supplementation Consider phosphate binders if renal impairment
	Hypophosphataemia	- Discuss with MO and Dietitian. Review medications that bind with phosphate - Manage any losses e.g. vomiting, Diarrhea malabsorption, refeeding syndrome - Consider supplementation
	Hypertriglyceridaemia and/or abnormal liver function	- Discuss with MO and Dietitian - Review feeding regime for overfeeding especially of carbohydrate
	Elevated Urea and/or Creatinine	Discuss with MO and Dietitian Check for dehydration or bleeding Consider changing formula

Refeeding syndrome - Severely malnourished - OR - Adapted to starvation	- At risk patient has one or more of: - Body Mass Index less than 16 kg/m² - Unintentional weight loss greater than 15% within the last 3–6 months - Little/no intake for more than 10 days - Low levels of potassium, phosphate or magnesium prior to feeding. Or - At risk patient has two or more of: - Body Mass Index less than 18.5 kg/m² - Unintentional weight loss greater than 10% within the last 3–6 months - Little/no intake for more than 5 days - Alcoholism or drugs e.g. insulin, chemotherapy, antacids or diuretics.	- Discuss identified refeeding risk with Dietitian as patients require less enteral nutrition to be provided initially until clinical and biochemical monitoring reveal no re-feeding problems - Enteral Feed: First 3 days 10 kcal/kg/day (other adults). Increase slowly to 15-20 kcal/kg/day (day 4-10). Done in conjunction with laboratory/clinical results - Consider providing intravenous thiamine, multivitamin and supplement electrolytes prior to commencing nutrition support - Monitor fluid balance - Check electrolytes 6 hours after commencing nutrition support and then at least daily during the first week - Supplement electrolytes as required. - Check electrolyte levels 3 days after stopping supplementation - Do not withhold all enteral nutrition to correct any electrolyte imbalance - If very little food for more than 5 days, introduce nutrition support at a maximum of 50% of requirements for the first 2 days
Food allergies	- Egg protein, soy protein, milk protein - Coeliac disease	Refer to Dietitian for specific advice
Food intolerances and avoidances	- Lactose, Amines, Salicylates, Glutamates FODMAPs - Pork, beef - Kosher, Halal, Vegan	Refer to Dietitian for specific advice